Validation matrix

Every reference case is an automated test (`cargo test`). A case passes only if the engine reproduces the published value within the stated tolerance. For Monte Carlo outputs the tolerance is ≈ 5 standard deviations of the engine's own dispersion, measured with `core/examples/mcm_dispersion.rs` (20 independent seeds, $M = 2 \times 10^6$), plus the rounding of the published value. Where published MCM values carry their own sampling noise, the primary reference is an analytic solution or an independent 2×10^8-trial PCG64 run (`validation/reference_values.py` → `reference_values.json`).

Test files: `core/tests/gum_examples.rs`, `core/tests/jcgm101_examples.rs`, `core/tests/eurachem_examples.rs`, `core/tests/topdown_examples.rs`, `core/tests/ea402_examples.rs`; unit tests in `core/src/*`.

Numerical building blocks

Item	Reference	Agreement	Test
erf, erfc, Φ^{-1}	SciPy	≤ 1e-12 relative	`special::tests`
Student t quantiles ν = 1…100, 2.5, 16.7	SciPy `stats.t.ppf`	≤ 1e-11 relative	`special::tests`
GUM Table G.2 ($t_{99}(16) = 2.92$ …)	GUM	2 decimals	`special::tests`
Wichmann–Hill 64-bit vs 32-bit form	JCGM 101 C.3.3.2/3	identical for 10^5 draws	`rng::tests`
t, gamma, normal generators	exact cdf/moments	statistical tests	`rng::tests`
Distribution moments (10 types)	JCGM 101 6.4	10^6 draws	`distribution::tests`
CTrap example 10.0 V ± 0.1 V → 0.060 V	JCGM 101 6.4.3	exact	`distribution::tests`
Numerical tolerance δ, examples 1–3	JCGM 101 7.9.2	exact	`mcm::tests`
Coverage interval indices	JCGM 101 7.7.2	exact	`mcm::tests`
χ^2 quantiles ν = 1…30	SciPy `stats.chi2.ppf`	≤ 1e-9 relative	`special::tests`
Rounding rules (MUPR Table 2, Nordtest round-up)	MUPR §12, Nordtest	exact strings	`report::tests`
Symbolic derivatives	central differences	≤ 1e-6 relative	`expr::tests`

Reference examples

ID	Source	Quantity	Published	Engine	Status
GUM-H1a	GUM H.1.6	u(d), ν_eff(d)	9.7 nm, 25.6	9.68 nm, 25.45 (25.6 with rounded u)	✅ (F1)

ID	Source	Quantity	Published	Engine	Status
GUM-H1b	GUM H.1, Table H.1	contributions	25, 9.7, 0, 0, 2.9, 16.6 nm	same	✅
GUM-H1c	GUM H.6, H.1.6	u_c, ν_{eff}, k_{99}, U_{99}	32 nm, 16.7, 2.92, 93 nm	31.67 nm, 16.76, 2.92, 92.5 nm	✅ (F1)
GUM-H1d	GUM H.1.7	u_c with 2nd-order terms	34 nm (terms 11.7, 1.7 nm)	33.8 nm (11.7, 1.7)	✅
S1-9.2.2	JCGM 101 Table 2	Gaussian additive	u = 2.00, [−3.92, 3.92]	agrees; GUF validated	✅
S1-9.2.3	JCGM 101 Table 3, Annex E	rectangular additive	analytic ±3.88	agrees	✅
S1-9.2.4	JCGM 101 Table 4	dominant rectangular	GUF ±19.9, MCM ±17.0; not validated (δ = 0.5)	agrees; validated only at ndig = 1	✅
S1-9.3	JCGM 101 Tables 6–7	mass calibration GUF1/GUF2/MCM	0.0539, 0.0750, 0.0754 mg	agrees; ref. 2×10^8: u = 0.075 484	✅ (F2)
S1-9.3a	JCGM 101 9.3.2.2	adaptive MCM δ/5	0.72×10^6 trials	converges, same results	✅
S1-9.4.2	JCGM 101 Table 8	comparison loss r = 0 (3 cases)	G1, G2, MCM	agrees with analytic χ^2	✅ (F2, F3)
S1-9.4.3	JCGM 101 Table 9	comparison loss r = 0.9 (3 cases)	MCM	agrees with 2×10^8 ref.	✅
S1-9.5	JCGM 101 Tables 10–11	gauge block (models 36, 37)	GUF 838, 32, [745, 931]; MCM 838, 36, [745, 932]	agrees	✅
EU-A1a	QUAM:2012 Table A1.3	Cd standard, u(V) = 0.07	1002.7 mg/L, u = 0.86 (0.9)	1002.6997, 0.8638	✅
EU-A1b	QUAM:2012 A1.4	volume sub-model	u(V) = 0.066 (0.07)	0.0665; u_c 0.835	✅ (F4)
EU-A2a	QUAM:2012 Tables A2.2–A2.3	NaOH standardisation	0.102136 mol/L, u = 0.000099, 5 contributions	identical	✅
EU-A2b	QUAM:2012 A2.4	sub-models (weighing, IUPAC molar mass, burette)	u(m) 0.13 mg, u(M) 0.0038, u(V) 0.013 mL, u_c 0.000099	0.122 mg, 0.0038, 0.0137 mL, 0.000101; MCM agrees	✅ (F10)
EU-A2c	QUAM:2012 Table A2.4	distribution choice	0.00011 / 0.000099 / 0.000085	not reproducible	⚠️ not tested (F10)
EU-A3a	QUAM:2012 Tables A3.2–A3.3	HCl titration (Kragten)	0.101387, u 0.00018, 7 contributions, (0.1014 ± 0.0004)	identical	✅

ID	Source	Quantity	Published	Engine	Status
EU-A3b	QUAM:2012 A3.1, A3.4, A3.6	summary table, triple determination, 25 °C, visual end-point	0.00016 (summary), 0.00016, 0.10149, 0.0003	0.00019 from summary inputs; others reproduced	✅ (F12)
EU-A4	QUAM:2012 Tables A4.2–A4.5, A4.6	pesticides: precision 0.27 (from raw duplicates), recovery t-test, binomial inhomogeneity (a)–(c), u_c/P 0.34, u 0.377	reproduced	✅ (F13)	
EU-A5a	QUAM:2012 Table A5.2, App. E.4	calibration from 15 raw points; u(c_0)	b_1 0.2410 (0.0050), b_0 0.0087 (0.0029), r 0.997, S 0.005486, u 0.018	identical	✅
EU-A5b	QUAM:2012 Tables A5.3–A5.4	Cd release budget and sub-models	r 0.015065, u 0.001465, u(V_L) 1.83, u(a_V) 0.19	0.001470 (exact), 1.83, u(a_V) 0.152	✅ (F14)
EU-A6	QUAM:2012 Tables A6.2–A6.5	crude fibre: s_R + drying allowance	0.31/0.4/0.6; U 0.62/0.8/0.12	U at 10 % = 1.2	✅ (F15)
EU-A7a	QUAM:2012 Tables A7.3–A7.8	lead by double IDMS, Eq. (5)	c_x 0.053738, u_c 0.00018 / 0.00041, ΣK_bias 37.6 %	reproduced	✅ (F16)
EU-A7b	QUAM:2012 Eq. (8), (12)	c_z, M(Pb)	0.092605 (0.000028), 207.2103	reproduced	✅

Top-down examples (relative values in %)

ID	Source	Quantity	Published	Engine	Status
NT-A5	Nordtest App. 5	NH_4-N duplicates: s_r < 30 µg/L; s′_r > 30 µg/L	0.44 µg/L; 3.8 %	0.441 (46 pairs); 3.82 (26 pairs)	✅ (F7)
NT-A6	Nordtest App. 6, §5.3	O_2 duplicates: pooled s, mean, s′; u(Rw)	0.0252 mg/L, 7.50, 0.34 %; 0.60 %	same	✅
NT-3	Nordtest §3.2, App. 4 (raw PT data)	NH_4-N: RMS_bias, ū(C_ref), u(bias), u_c, U	2.26, 1.52, 2.73, 3.20, 6.4 → 7 %	same	✅
NT-6.1a	Nordtest §6.1	one CRM	u(C_ref) 2.16, u(bias) 4.1 %	2.17, 4.15 % (4.1 with printed 2.16)	✅ (F5)
NT-6.1b	Nordtest §6.1	three CRMs	2.53, 1.92, 3.2 %	same	✅
NT-6.2	Nordtest §6.2	proficiency tests	1.8; 2.7/2.6; 4.6; 5.3 %	1.81; 2.60; 4.60; 5.28	✅ (F6)
NT-6.3	Nordtest §6.3	recovery	3.44, 0.76, 1.0, 3.6 %	same	✅
NT-8.2	Nordtest §8.2, App. 7 (raw data)	BOD: mean, s, s′; Examples A and B	214.8, 5.6, 2.6 %; 4.5/5.2/11 %; 1.69/3.76/4.12/4.87/10 %	same (RMS 3.77 unrounded)	✅ (F8)
NT-8.3	Nordtest §8.3	PCB Examples C and D	7.22/10.8/22 %; 7.6/8.1/11.4/23 %	same	✅ (F8)

ID	Source	Quantity	Published	Engine	Status
MUPR-E2	MUPR Example 2	mean recovery, 2 CRMs + PT	$\bar{R}$ 1.008, u 0.03705 (3 terms)	same	✅
MUPR-E3	MUPR Example 3	spiked samples with native analyte	96.2 %, 10.02 %; 1.5 % (A only)	same	✅
MUPR-B1	MUPR Example B1	nitrate by IC: $\bar{R}$, u_$\bar{R}$, Eq. 13, limits, U(I), U(II), contributions, B1.16, B1.18, reporting	97.2 %, 0.0285, 0.982 ≤ 2.093, 0.794/1.15, 0.729/1.21, 0.0676, 0.185c, 94.1/4.8/1.1 %, 82.9/9.5/2.1/5.6 %, 7.13, 0.57 ≤ 1.27, (0.261 ± 0.068), (50.0 ± 9.3)	same (0.963 and 0.0849 unrounded)	✅ (F9)
MUPR-12	MUPR §12, Table 2	rounding of reported results	(0.156 ± 0.012), (5360 ± 240), (5360 ± 260)	same	✅
MUPR-B2	MUPR B2 (raw Tables B2.1, B2.3)	pooled precision, two recoveries, compatibility, overall recovery, U, replicates	0.0285, 7.73 %, 0.048 ≤ 0.134, 99.6 %, 2.24 %, 0.169c, 7.41, 7.06	same	✅ (F17)
MUPR-B3	MUPR B3	regression precision model; U with F = 100; k = t(5)	0.0233 + 0.0128c (r 0.7266), 7.28, 0.137c	intercept 0.02324	✅ (F17)
MUPR-B4	MUPR B4	As in sediments, one CRM	4.52 %, 0.0381, 2.24 ≤ 2.26, (16.0 ± 1.9), U_tg	same	✅
MUPR-B5	MUPR B5	CRM + 2 spiked samples, matrix variability	$\bar{R}$ 0.9498, s(R) 0.0732, u 0.0800 / 0.0323, pairwise 0.275/1.32/1.50	same	✅
MUPR-B6	MUPR B6 (raw Tables B6.1–B6.3)	heterogeneity from 30 duplicates, 22 PT recoveries, χ^2, intervals	7.34 %, 0.2157, 3.00 %, 0.5828, 7.808 %, $\bar{R}$ 0.9174, u 0.01585 / 0.04872, χ^2 8.80, 0.160, 0.1157, 21.5, 29.7	same (χ^2 8.70 with weighted mean)	✅ (F17)
MUPR-C1	MUPR App. C, Example C1	allowance for an accepted recovery interval	Δa 27.87 %, u_a 8.1 %	8.05 %	✅ (F17)

Calibration examples (EA-4/02 M:2022)

ID	Example	Published	Engine	Status
EA-E1	Table E.1 coverage factors (26 values)	13.97 … 2.00	$t_{95.45}(\nu)$ agrees	✅
EA-S2	weight 10 kg	29.2 mg, U 58 mg	29.26 (29.2 from rounded contributions)	✅ (F18)
EA-S3	resistor 10 kΩ (Type A from 5 ratios)	10 000.178 Ω, 8.33 mΩ, U 17 mΩ	same	✅
EA-S4	gauge block 50 mm (second-order δα·θ)	49.999 926 mm, 34.3 nm, U 69 nm	same	✅ (F19)
EA-S5	thermocouple 1000 °C (two budgets)	0.641 °C; 36 229 µV, 25.0 µV	same (δt term 24.65)	✅ (F18)
EA-S6	power sensor (U-shaped mismatch)	0.933, 0.01623, U 0.032, ν_eff ≈ 310	same	✅ (F18)
EA-S7	step attenuator	30.043 dB, 0.0223 dB, U 0.045 dB	0.0222 dB	✅ (F18)
EA-S9	DMM 100 V, rectangular output	0.030 V, k 1.65, (0.10 ± 0.05) V	same	✅

ID	Example	Published	Engine	Status
EA-S10	vernier calliper, trapezoid β = 0.33	32 µm, k 1.83, (0.10 ± 0.06) mm	same	✅
EA-S11	temperature block, β = 0.43	164 mK, k 1.81, (180.1 ± 0.3) °C	k = 1.80	✅ (F18)
EA-S12	water meter (ν_eff = 10)	199.93 l, 0.109 l; 0.68e-3; 0.91e-3, k 2.28, U 0.002	same	✅
EA-S13	ring gauge 90 mm (temperature products)	0.15 µm sub-budget, 0.411 µm, U 0.9 µm	0.148, 0.41, 0.82 → 0.9 only with round-up	✅ (F18)

NIST Technical Note 1900 (2015) — library projects

ID	Example	Published	Engine	Status
TN-E1	weighing (linear, Gaussian)	50.284 g, u 0.0071 g, [50.270, 50.298] g	same; GUM validated	✅
TN-E2	surface temperature (Type A, ν = 21)	25.59 °C, u 0.872 °C, k 2.08, (23.8, 27.4) °C	same (GUM and MCM)	✅
TN-E3	falling-ball viscometer (skewed)	MCM 5.82, u 1.11, [4.05, 8.39] mPa s; “Gauss” 1.11	MCM same; 1st-order GUM 1.027, 2nd-order 1.10	✅ (F29)
TN-E4	Pitot tube (power law)	40.64 m/s, u 0.25 m/s, [40.17, 41.13]	57.48 m/s, u 0.348 m/s (= printed × √2)	✅ (F28)
TN-E8	M_r(CO_2), two rectangular inputs	44.0094, u 0.0007, exact [44.0080, 44.0108]	same (exact 44.008045, 44.010755)	✅ (F32)
TN-E9	cadmium standard (QUAM A1, rectangular P, triangular V)	1002.7 mg/L, u ≈ 0.9, [1001.0, 1004.4]	0.864; same interval	✅ (F30)
TN-E11	step attenuator (EA S7, arcsine δL_M, δL_D, δL_K; t_3 L_S)	30.043 dB, u 0.0224 dB, [30.006, 30.081]	same (MCM only)	✅ (F31)
TN-E13	thermal expansion, Gaussian / t_3 inputs, 95 % / 99 %	1.73e-5, u 0.18e-5 K^{-1}; (1.38, 2.07), (1.27, 2.18), (1.40, 2.05), (1.15, 2.30)	same (4 projects)	✅ (F31)

NIST Uncertainty Machine manual v1.4 (2020) and GUM H.2 — library projects

ID	Example	Published	Engine	Status
NUM-3	y = ab/c, triangular divisor	Gauss 28.8, u 8.7; MCM 32.2, sd 13, (18.2, 67)	same	✅ (F33)
NUM-8	end gauge, Gaussian inputs	Gauss u 31.7 nm; MCM sd 33.9 nm, 99 % (50 000 749, 50 000 927)	same; zero-sensitivity warning 13 %	✅
NUM-10	GUM H.2 R with rounded Table H.2 values	Gauss 0.07; MCM sd 0.0699, (127.595, 127.869) Ω	0.0700; same	✅ (F34)
NUM-11	Stefan–Boltzmann constant (CODATA 2010)	5.670 3725e-8, u 2.05e-13	same	✅
GUM-H.2	R, X, Z from the raw sets of Table H.2	0.071, 0.295, 0.236 Ω	0.0711, 0.2956, 0.2363	✅ (F34)
NUM-7, NUM-9	thermal expansion (t_3), viscosity	= TN 1900 E13, E3	see above	✅ (F29, F35)

ISO 13528:2022 Annex E — library projects

ID	Example	Published	Engine	Status
ISO13528-E.2	homogeneity and stability, As in chocolate	$\bar{x}$ 0.18715, s_x 0.00398, s_w 0.00556, s_s 0.00060, 0.3σ_pt 0.00842; Δ 0.00660	same	✅
ISO13528-E.3	atrazine, 34 results, Algorithm A	x* 0.2570, s* 0.0395 (6 iterations, x* ± δ of each), u 0.0085; median 0.2620, MADe 0.0386, nIQR 0.0402	same	✅
ISO13528-E.3	median and nIQR	0.2620, 0.0402, u 0.0086	same	✅
ISO13528-E.4	mercury, reference value, 21 laboratories	D %, PA, z, z′, ζ, E_n (126 values)	same	✅ (flags: F36)
ISO13528-B.1	factors F_1, F_2	Table B.1 (g = 7 … 20)	from χ^2 and F quantiles	✅
ISO13528-E.1	23 results, 5 censored, Algorithm A	‘<’ ignored 26.01/7.23; deleted 26.81/5.29; 0.5 × limit 23.95/8.60	same; 0.5 × limit converged 23.959/8.600	✅ (F44)
ISO13528-E.7	robust mean of E.4 vs reference value	x* 0.03161, s* 0.0164 (21 results)	same	✅
ISO13528-E.5	reference value from a CRM, 20 paired samples	d 1.73, s 1.07, u(d) 0.24; x_pt 23.35, u 0.35	same	✅

ISO 21748:2017 Annex C and ISO 5725-3:2023 Annex J — library projects

ID	Example	Published	Engine	Status
ISO21748-C.1	CO in emissions, s_R = 0.28 g/km	U 0.56 g/km	same	✅
ISO21748-C.2	meat content (combined result)	95.6 %, u 1.98 %, U 4.0 %	95.6, 1.99, 4.0	✅
ISO21748-C.3	aerobic plate count, s′_R with s_l (7.3.2) + 3 %	shrimp 7.2/7.8, vegetables 8.4/8.9, flour 5.5/6.4 %	flour 6.29 %	✅ (F39)
ISO21748-C.4	crude fibre, s_R by range + drying	u 0.31, 0.41, 0.59 %; U 0.6, 0.8, 1.2 %	same	✅ (F38)
ISO5725-3-J.1	carbon in steel, 27 day-pairs	s_I(TO) 2.87 × 10^{-3} %	same	✅
ISO5725-3-J.2	vanadium, staggered-nested, levels 1–2	SS, MS, s_0^2, $s_{(1)}^2$, s_r, s_I(T), s_R	same	✅
ISO5725-3:1994-D.2	vanadium, levels 3–6 from the 1994 data (core test)	Table D.5 = J.5: s_r, s_I(T), s_R	same at levels 3–5; level 6 s_R 16.331 against 15.962 (negative $s_{(1)}^2$ kept, D41; the printed s_R equals the REML estimate, a difference of method, D108)	✅ (F40, D41, D108)

JCGM 102:2011 — library projects and core tests

ID	Example	Published	Engine	Status
JCGM102-T1–T2	coverage factors, m = 1 … 50	k_p (Table 1), k_q (Table 2)	same	✅ (core)
JCGM102-6.5.2.3	ellipse and rectangle areas	26.6, 28.4; 11.8, 40.1 unit²	26.6, 28.4; 11.8, 40.2 (40.1 with k_q = 2.24)	✅ (F50)
JCGM102-6.5.4	coverage region for an expectation	$F_{2,10}(0.95) = 4.10$	same	✅ (core)
JCGM102-7.7	L, k_p, smallest region	L (transposed, F50); k_p 2.33 and 10.6 unit² from 1 000 draws; smallest region 9.4 unit² (10^6 draws)	k_p 2.27, 10.1 unit² (10^6 draws); smallest region 9.36 unit²	✅
JCGM102-9.2.2	additive model, Gaussian	Table 3: GUF 1.414, 0.500, 2.45, 2.24; MCM 2.45, 2.21; validated	same	✅
JCGM102-9.2.3	rectangular common effect	Table 4: MCM k_p 2.38, k_q 2.15; regions not validated	same	✅
JCGM102-9.2.4	dominant rectangular effect	Table 5: GUF 3.162, 0.900; MCM 2.28, 1.87	same	✅
JCGM102-9.3	polar co-ordinates, six cases	Tables 6, 7 (MCM, M = 10^7)	same (M = 10^6)	✅
JCGM102-9.4	resistance, reactance, impedance	Table 11: GUF, Alt. GUF, MCM (t_3)	same; MCM within the seed-to-seed spread of t_3	✅ (F51)
JCGM102-9.5.2	resistance thermometer, implicit	θ 20.023 2 °C, u 0.004 5 °C	same	✅
JCGM102-9.5.3	ten temperatures, implicit	Tables 15, 16	same	✅

ISO 5725-2:2019 — library projects and core tests

ID	Example	Published	Engine	Status
ISO5725-2-T5–T8	critical values and indicators	Table 5 (p = 8, 9; n = 2, 3), Table 6 (p = 8, 9, 15, 16, 40), Tables 7, 8 (p = 9)	D.1, D.2, D.5, D.6 within the printed digits (Table 6 single: ≤ 0.001); D.3 within ±0.003 of Table 6 for p = 6 … 40	✅ (core)
ISO5725-2-C.1	sulfur in coal, 8 labs × 4 levels, unequal n	Cochran C (C.1–C.4), Grubbs (Table C.4), $\hat{m}$, s_r, s_R (Table C.5)	same from Forms rounded to 3 decimals	✅ (F47)
ISO5725-2-C.1-REML	REML	Table C.6	same	✅
ISO5725-2-C.2	softening point of pitch, 16 labs, n = 2, missing cells	Cochran (Table C.10), Grubbs (Table C.11), $\hat{m}$, s_r, s_R (Table C.12)	same; s_R at level 4 1.9175 against 1.915	✅ (F48)
ISO5725-2-C.2-REML	REML	Table C.13	same	✅

ID	Example	Published	Engine	Status
ISO5725-2-C.3	creosote oil, outlying laboratory	Cochran 0.667, 0.636; Grubbs (Table C.17); m̂, s_r, s_R (Table C.18); relationships I–IV (Tables 1–4)	same; IV slope 0.770 against 0.772	✅ (F49)
ISO5725-2-C.3-REML	REML	Table C.19	same	✅
ISO5725-2-A.1	A_r, A_R	Table A.1 (5 entries)	same	✅ (core)

ISO 6143:2025 Annex B — library projects

Reference: B_LEAST output printed in the standard. Independent checks with ODRPACK (scipy.odr 1.18, `cov_beta`) and a Monte Carlo simulation of the whole calibration (4000 refits).

ID	Example	Published	Engine	Status
ISO6143-B.2.2	NO by UV, linear, 3 points	b, u(b), cov, S_res 0.6743, Γ 0.568, weighted deviations (Table B.4), 3 contents with u and covariances (Table B.6)	same	✅
ISO6143-B.2.3	N_2 by GC-TCD, linear, 8 points	b, u(b), cov, S_res 6.0455, Γ 1.6266, Table B.8	same	✅
ISO6143-B.2.3	quadratic	b, u(b), 3 covariances, S_res 1.3964, Γ 0.8664, Table B.9	same	✅
ISO6143-B.2.3	linear with u(x_4, x_7), u(x_5, x_8)	u(b), cov, Table B.10	same	✅
ISO6143-B.2.4	NO by NDIR, 12 points, linear (rejected)	S_res 272.6392, Γ 6.8352	same	✅
ISO6143-B.2.4	power function	b, u(b) and covariances within 0.3 % (ODRPACK 0.9 %), S_res 8.3804, Γ 1.1594, x 5.3456, u 0.014169	same	✅
ISO6143-B.2.4	exponential function	b, S_res 0.6572, Γ 0.3529, x 5.3357; u(b) 8.14×10^{-2}, 7.40×10^{-2}, 3.71×10^{-11}; u(x) 0.013280	b, S_res, Γ, x same; u(b) 3.18, 3.17, 1.27×10^{-6} and u(x) 0.014246 as ODRPACK (MC: 3.13, 3.12, 1.24×10^{-6}; 0.01406)	✅ (F42)
ISO6143-6.2.4	drift-test critical values	$2\sqrt{2}$ u for n = 10 (Note); Formula (12) $2\sqrt{(1 + 20/n)}$ u	$2\sqrt{(1 + 10/n)}$ u, $2\sqrt{(20/n)}$ u	✅ (F41)

Palma et al. (2019), marine sediments — library projects

Two-interval top-down models (LOD … 2w_LOQ absolute, above 2w_LOQ relative) combined from the components of Table 9 (raw data only in the electronic supplement).

ID	Element	Published u_c⟨I⟩ / u′_c⟨II⟩	Engine	χ_ME⟨I⟩ / ⟨II⟩ published / engine	Status
Palma-Cr	Cr (OSPAR)	1.05 mg/kg / 10.5 %	1.049 / 10.49	60.8 / 60.9 — 60.8 / 60.8	✅
Palma-Cu	Cu (OSPAR)	0.612 / 6.13	0.6128 / 6.123	64.7 / 64.5 — 64.7 / 64.6	✅ (rounded components)
Palma-Li	Li (OSPAR)	0.127 / 12.7	0.1267 / 12.67	62.3 / 62.4 — 62.3 / 62.3	✅
Palma-Mn	Mn (OSPAR)	0.920 / 9.20	0.9198 / 9.198	42.6 / 42.6 — 42.6 / 42.6	✅
Palma-Ni	Ni (EPA 3051A)	1.25 / 8.44	1.253 / 8.442	7.9 / 7.7 — 7.9 / 7.7	✅
Palma-Pb	Pb (EPA 3051A)	1.90 / 10.1 (10.08 in Eq. 21)	1.895 / 10.08	65.2 / 57.6 — 65.2 / 57.6	✅
Palma-Zn	Zn (OSPAR)	0.365 / 9.13	0.3655 / 9.132	78.7 / 78.6 — 78.6 / 78.7	✅

Bettencourt da Silva (2026), MUPR spreadsheet — library projects

Fluoranthene in river sediments, Supplementary Material 06: 20 mean recoveries with u(R_k) from the sheet “Trueness”, two-interval precision model, s(R) added (χ^2 test failed), k = 2.093 (ν = 19). Expected values are recomputed by an independent script. The spreadsheet’s own uncertainties differ for the reasons of F72 and F73.

ID	Mean recovery	χ^2 / s(R)	Spreadsheet	Engine (0.1 / 10 mg/kg)	Status
Bet-weighted	$\bar{R}$ = 74.887 % (Eq. 16.1), u = 8.812 %	40.973 / 8.755 %	$\bar{R}$ 74.887 %, χ^2 40.973, s 8.755 %; U 0.161 / 3.857	(0.1335 ± 0.2127) / (13.353 ± 3.575)	✅ $\bar{R}$, χ^2, s; U by design (F72, F73)
Bet-arithmetic	$\bar{R}$ = 76.307 % (Eq. 8–12), u = 9.629 %	40.973 / 8.755 %	—	(0.1311 ± 0.2091) / (13.105 ± 3.724)	✅ recomputed

JCGM 106:2012 — conformity assessment (library projects and core tests)

ID	Example	Published	Engine	Status
JCGM106-7.3-E1	Zener diode, V_b ≤ −5.40 V	p_c 0.92	0.919	✅
JCGM106-7.3-E2	bursting strength, B ≥ 490 kPa	p_c 0.99	0.989	✅
JCGM106-7.4	motor oil, two-sided	p_c 0.66 (0.93 − 0.27)	0.663	✅
JCGM106-7.7.5	C_m = 1, 95 % locus	0.45 ≤ ỹ ≤ 0.55	0.4491 … 0.5509	✅ (core)
JCGM106-8.3.2	w = U = 2u (ISO 14253-1)	false acceptance ≤ 2.3 %	2.28 %	✅ (core)
JCGM106-8.3.3-E1	speed limit, u = 2 % v, 99.9 %	v_max ≈ 107 km/h	106.59	✅
JCGM106-8.3.3-E2	nandrolone, t_9, 95 %	A = 2.37 µg/L	2.367	✅
JCGM106-9.5.3	precision resistors	p_c 90 %, R_C 1 %, R_P 7 %; 83/7/9/1 per 100	90.4, 0.99, 6.90 %; 83.5/6.9/8.6/1.0	✅
JCGM106-9.5.4	ball bearings, gamma prior	4.2 % non-conforming; r ≈ 0.65, A ≈ 1.7 µm, R_P ≈ 7.5 %	4.24 %; 0.656, 1.672, 7.55 %	✅ (F45)

ID	Example	Published	Engine	Status
JCGM106-9.5.6	u_0 = T/6, w = 0, C_m = 2 and 10	R_C ≈ 0.1/0.04 %, R_P ≈ 1.5/0.07 %	0.098/0.041 %, 1.47/0.072 %	✅ (core)

Nordtest TR 569 ed. 6 (2026) — control charts (library projects and core tests)

ID	Example	Published	Engine	Status
TR569-T4	range factors, n = 2 … 4	d_2 1.128; D_WL 2.833, 3.470, 3.818; D_AL 3.686, 4.358, 4.698	d_2, d_3 by numerical integration (n = 2 … 10); D_WL 2.8334, 3.4693, 3.8184; D_AL 3.6859, 4.3577, 4.6982	✅ (core, F54)
TR569-T1	Zn, 60 raw values, X-chart	Figure 1: no value out of control	$\bar{x}$ 60.278, s 2.598; WL 55.08/65.47, AL 52.48/68.07; 3 values in the warning zone, 0 out of control; t = 0.83 against 60.0	✅
TR569-7-X1–X3	X-chart cases 1–3	54.5/63.9, 52.1/66.3; 53.3/65.1, 50.3/68.1; 54.0/66.0, 51.0/69.0 µg/L	same	✅
TR569-7-R1–R2	r%-chart cases 1–2	WL 1.0 %, AL 1.3 %; s_r = r/2.8 = 0.357 %	1.01/1.31 %; 0.357 %	✅ (F52)
TR569-E1	Ni by XRF, target from U = 4 %	s 0.0458; 4.49/4.67, 4.44/4.72	same	✅
TR569-E2	Co by XRF, target from LOQ	0.0748/0.0788, 0.0738/0.0798	same	✅
TR569-E3	NH_4-N, X- and R-chart	18.95/21.03, 18.43/21.55; s 0.496, WL 1.40, AL 1.83	same (s = 0.4954 from 0.559/1.128379)	✅
TR569-E4	Pb by ICP-MS	0.278/0.310, 0.270/0.318	same	✅
TR569-E5	As in CRM, tolerance ± 15 %	AL 15.3/20.7	same	✅
TR569-E6	total P, r%-chart	CV 1.67; WL 4.73, AL 6.16 %	same with the rounded CV (4.72/6.14 % unrounded)	✅ (F54)
TR569-E7	β-HCH, target 15 % around the reference value	11.2/20.8, 8.8/23.2	same	✅
TR569-E8	Cu by ICP-OES: preliminary and new limits, review	0.92/1.19, 0.85/1.26; F 1.563 (crit 1.67), s_p 0.07545, t 1.012 (crit 1.98); 0.88/1.21, 0.80/1.29; R-chart 0.28/0.36, F 1.037	same; F_crit 1.677 exact; s_p 0.07544; F (R) 1.0380	✅ (F54)
TR569-E9	Zn blank	−0.051/0.129, −0.096/0.174	same	✅
TR569-E10	B, target 10 % around the reference value	4/6, 3.5/6.5	same	✅
TR569-E11	micropipette, tolerance ± 0.01 mL	AL 0.99/1.01	same	✅
TR569-E12	3 × 8 replicate control results	s_r 0.15 (Eq. 10), s_Rw 0.270 (simplified), 0.272 (ANOVA; footnote: s_r 0.1541, s_I 0.2246)	0.1541, 0.2705, 0.2724 (s_r 0.1541, s_I 0.2247)	✅ (F53)
TR569-9	daily and statistical-control rules	Chapter 9	core tests: action limit, two out of three (D43), trend of seven, ten of eleven (Example 4: twelve values on one side)	✅ (core)

UKAS M3003 ed. 6 (2024) — library projects and core tests

ID	Example	Published	Engine	Status
M3003-B	t-table for p = 68.27, 95, 95.45, 99.73 %; B.10	table; ν_eff 21.1, k 2.13	same	✅ (core)
M3003-C7	k(95.45 %) for rectangular ⊛ normal, u_N/u_R = 0 … 2.5	26 values, 1.65 … 2.00	all within ±0.005	✅ (core)
M3003-C5	DVM with dominant resolution	u_c 0.305 mV, k 1.77, U 0.54 mV	same; MCM interval ±0.54 mV, validated	✅
M3003-K1	10 kΩ resistor	u_c 0.445 ppm; 10 000.1050 Ω ± 0.0089 Ω	same	✅
M3003-K2	power sensor, 18 GHz	u_c 1.69 %, U 3.39 %	same	✅
M3003-K3	30 dB attenuator	u_c 0.0245 dB, U 0.0491 dB	same (0.024 57); with E5 mismatch 0.0202 dB	✅ (F57)
M3003-K4	10 kg weight	u_c 24.56 mg, U 49.12 mg	same	✅ (F55)
M3003-K5	weighing machine, 200 g	u_c 0.150 mg, U 0.300 mg, +0.10 mg	0.153 mg, 0.306 mg, −0.10 mg	❌ printed (F56)
M3003-K6	10 mm gauge block	u_c 40.7 nm, U 81.5 nm	same; with the model's temperature terms 41.0 nm	✅ (F58)
M3003-K7	N-type thermocouple, 1000 °C	u_c(t) 0.641 °C; u_c(V) 25.9 µV, U 51.9 µV, 36 230 µV	0.641; 25.84, 51.7, 36 228.8	✅ (F59)
M3003-K8	DPI at 2 MPa	u_c 43.0 ppm, U 86.0 ppm, +17 ppm	same	✅ (F60)
M3003-K9	flowrate at standard conditions	12.53 L/s, u_c 0.275, ν_eff 130, U 0.55	same	✅
M3003-K10	sheet thickness	1.514 mm, u_c 0.009 63 mm, ν_eff 50, U 0.019 mm	same	✅
M3003-K11	PT100 probe; gas temperature	u_c 0.042 25 °C; 22.97 °C, u_c 0.086 38 °C	same; 0.086 35 °C	✅
CG18-H1	220 g / 0.1 mg balance: zero, 220 g (5d, 5e), 100 g and 220 g with buoyancy correction	u(E) 0.000 118, 0.001 971, 0.000 491, 0.000 149, 0.000 226 g; k 2.87, 2.00, 2.00, 2.25, 2.05	same; k = 2.04 exact at ν = 62 (guide uses GUM Table G.2)	✅

ID	Example	Published	Engine	Status
CG18-H2	60 kg multi-interval balance (2/5/10 g): zero, 10 kg, 60 kg (5d, 5e, option 2), adjusted 60 kg (5c)	u(E) 1.238, 1.552, 5.978, 5.948, 5.941, 5.739 g; k 2.52, 2.17, 2.05, 2.05, 2.05, 2.025	1.238, 1.552; 5.973, 5.944, 5.938, 5.725 (mpe 0.90 g for 3 × 20 kg F2); k exact 2.03, 2.01	✅ (F68)
CG18-H3	30 t weighbridge, substitution loads: zero, 10 t, 15 t, 30 t	u(E) 6.75, 7.98, 14.60, 23.85 kg; ν_eff 5, 9, 109, 783; U 18, 19, 29, 48 kg	6.75, 7.98, 14.60, 23.86; ν_eff 5, 9.8, 37, 157; U 17.9, 18.5, 30.3, 48.1	✅ u(E); ν_eff differs (F69)
CG18-H4	error function E = a_1I, 400 g balance, s_m = 0, 0.05, 0.25 mg	a_1 0.00083/0.00084; U(â) $5.109\cdot10^{-8}$, $5.637\cdot10^{-8}$, $1.745\cdot10^{-7}$; χ^2 12.5, 7.3; u(E_appr) Tables H4.6, H4.8; 2 points fail C2.2-2b at s_m = 0.05	same; minimal s_m 0.044 mg (C2.2-2a), 0.22 mg (C2.2-2b)	✅ (F70)
CG18-H1.4/A	weighing result in use, 220 g balance (5e fit, built-in adjustment, ΔT = 3 K)	a_1 $6.709\cdot10^{-6}$, u(a_1) $1.242\cdot10^{-6}$; α^2 $1.467\cdot10^{-8}$ g², β^2 $8.390\cdot10^{-12}$; U_gl = $2.422\cdot10^{-4}$ g + $1.150\cdot10^{-5}$R; R_min,SF 0.0729 g (1 %, SF 3)	$6.710\cdot10^{-6}$, $1.242\cdot10^{-6}$; $1.466\cdot10^{-8}$, $8.389\cdot10^{-12}$; $2.422\cdot10^{-4}$ + $1.150\cdot10^{-5}$R; 0.07292 g	✅
CG18-H1.4/B	same, adjusted before calibration (5c fit)	a_1 $3.895\cdot10^{-7}$ (sign omitted), u(a_1) $6.337\cdot10^{-7}$; tare $5.774\cdot10^{-7}$; β^2 $6.433\cdot10^{-12}$; U_gl slope $4.479\cdot10^{-6}$; R_min,SF 0.0727 g	$-3.898\cdot10^{-7}$, $6.332\cdot10^{-7}$; tare $1.155\cdot10^{-6}$ (7.4.4-5); $7.432\cdot10^{-12}$; $4.851\cdot10^{-6}$; 0.07277 g	✅ except tare (F76)
CG18-H2.4/A	weighing result in use, 60 kg multi-interval (5d fit), three PWI	a_1 $1.717\cdot10^{-4}$ (sign omitted), u $6.459\cdot10^{-5}$; α^2 1.867/9.917/16.167 g²; β^2 $4.589\cdot10^{-8}$; slopes $2.574/3.434/3.923\cdot10^{-4}$; U_gl 2.733 + $4.291\cdot10^{-4}$R, 10.190 + $5.151\cdot10^{-4}$(R − 12 kg), 20.311 + $5.641\cdot10^{-4}$(R − 30 kg); R_min,SF 598 g	same; U(W) at 12 and 30 kg 8.129 and 15.16 g (printed 10.190, 20.311)	✅ except two intercepts (F75)
CG18-H2.4/B	same, adjusted before calibration (5c fit)	a_1 0, u $6.043\cdot10^{-5}$; α^2 1.467/7.417/13.667 g²; β^2 $2.449\cdot10^{-8}$; U_gl 2.422 + $1.706\cdot10^{-4}$R, 6.616 + $2.355\cdot10^{-4}$(R − 12 kg), 11.951 + $2.744\cdot10^{-4}$(R − 30 kg); R_min,SF 502 g	same; 501.5 g	✅
GUM5-5	GUM-5 Example 5, GHG inventory (agriculture A4ci, UK 2018), independent and shared activity data	E 4282.04 kt, u 835.16 kt; Table 5.3 contributions	E 4282.14 (F_1 rounded), u 835.16; contributions to 10^{-4} kt; u_cor − u_ind as (5.9)	✅ (F84 class a/b)
GUM5-7	JCGM GUM-5:2026 Example 7, mass calibration (= JCGM 101 9.3)	GUF1 1.2340, 0.0539, [1.1285, 1.3395] mg; GUF2 0.0750, [1.0870, 1.3810]; MCM 0.0755	same; second-order estimate term 0	✅
GUM5-16	JCGM GUM-5:2026 Example 16, comparison loss, r = 0 and 0.9	Table 16.1 GUF2 δy 50/150/2550, u 50/112/502; Table 16.2 analytic u 67/121/505 (×10^{-6})	δy with the Amd.1 term 50/150/2550; u 50/111.8/502.5; correlated GUF2 (D57) 67.27/120.52/504.50	✅ (D56, D57)
GUM5-1	GUM-5 Example 1, pH by interpolation (25 °C; correlated standards; 23.7 °C)	7.0002, u 0.0041; u 0.0051; 7.0043 (printed 7.0109), u 0.0041	7.0002, 0.0041; 0.0051; 7.0043, u 0.0032 independent / 0.0041 same-standard correlated	✅ except F78
GUM5-2	GUM-5 Example 2, benzo[a]pyrene by internal standard (correlated pairs)	0.014 ng, u^2 $4.1\cdot10^{-6}$ ng², ν_eff 5.2 (G.2b), 4.8 (eq. 2.3); MCM [≈ 0, 0.032] ng	0.01403, $4.08\cdot10^{-6}$, 5.20, 4.81; MCM script [$1\cdot10^{-7}$, 0.0321]; ensemble ν_eff 2.11	✅ except F79

ID	Example	Published	Engine	Status
GUM5-3	GUM-5 Example 3, molar mass of glucose	180.1557, u 0.0038	same	✅
GUM5-4	GUM-5 Example 4, gravimetric CO_2/N_2 mixture (ISO 6142-1), 8 outputs	Table 4.6: Ar 4.0 ± 2.4 … CO_2 200 345 ± 5, H_2O 0.379, N_2 799 649 ± 5; r(CO_2, N_2) −0.86, r(Ar, N_2) −0.50; R_M (4.3.4.6)	same except CO_2 ± 3.96, H_2O 0.389, r −0.85 and −0.55; R_M identical	✅ except F82
GUM5-9	GUM-5 Example 9, torque slope by OLS/WLS (GUF) and Bayes	OLS 1.0107 ± 0.0015 [1.0055, 1.0158], ν 2.7; WLS 1.0085 ± 0.0008 [1.0069, 1.0101], ν 14; NI-Bayes 1.0092 ± 0.0011; Inf-Bayes 1.0091 ± 0.0009	same; WLS ν 13.31; Bayes by quadrature (script)	✅ except F83
GUM5-10	GUM-5 Example 10, conformity of TSPM with a lognormal prior	R_C 0.5 %, R_P 0.9 % (A_U = T_U); 0.002 %, 5.8 % (0.17); 1.7 %, < 0.1 % (0.22); specific R_P ≈ 11 % at 0.225	printed form: 0.498/0.909, 0.0020/5.78, 1.69/0.115 %; conditional h: 0.579/0.739, 0.0103/5.37, 1.80/0.055 %; posterior 11.33 %	✅ except F80
GUM5-11	GUM-5 Example 11, tumour area from boundary pixels (Fourier series, MCM)	Table 11.1 areas; MCM n = 6 65.4 ± 1.6 (11.6.1: 65.0); n = 5 65.3, n = 7 65.0; AIC/AICc/BIC minimal at n = 2	data confirmed by EMUE E5.1.5; areas ±0.1 (EMUE Table E5.1.1 likewise); MCM 65.45 ± 1.55 (EMUE 65.5 ± 1.6); IC with $\chi^2 = S/u_0^2$: n = 6/4/4; unweighted S gives n = 2	script only; F85 (+ addendum)
GUM5-12	GUM-5 Example 12, between-bottle homogeneity, Bayesian τ	MS $3.25\cdot10^{-7}/4.90\cdot10^{-7}$; τ 0.00014 ± 0.00011 [0.00001, 0.00040]; σ 0.00069; HPD τ/μ [0, 0.0008], σ/μ [0.0013, 0.0020]; median 1.1, mode 0.3 μmol/mol	same; τ $1.356\cdot10^{-4}$ ± $1.07\cdot10^{-4}$ [$5.2\cdot10^{-6}$, $3.97\cdot10^{-4}$]; median 1.13; mode 0; u*_bb 1.48 μmol/mol	✅ except F81
GUM5-14	GUM-5 Example 14, ^{207}Bi decay correction	10.810 kBq, u 0.054 kBq; Table 14.1	same	✅
GUM5-15	GUM-5 Example 15, wire rope breaking force	10 005.7 kN, u 9.9, U 19.4 kN; MCM u 10.1 kN	GUF same; MCM 9.93 kN	✅ GUF; MCM differs (F77)
CG19-E1	1000 mL flask (Tanaka density)	V_{20} 999.628 mL, u 0.017 mL, ν_eff 19, k 2.15, U 0.036 mL	999.622; 0.0169; 19.5; 2.14; 0.036	✅ (F62)
CG19-E2	100 mL pycnometer	100.014 mL, u 0.89 μL, ν_eff 13, k 2.21, U 2.0 μL	100.0142; 0.90 μL; 12.9; 2.23 (G.6.4 truncation); 2.0 μL	✅ (F62)
M3003-L3	DMM, 1 V range	0.95 V: u_c 5.37 μV; 500 mV: U 6.2 μV; 1 V: k 1.898	same	✅ (F61)

Eurachem UfS:2019 — uncertainty from sampling (library projects and core tests)

ID	Example	Published	Engine	Status
UFS-A1	nitrate in lettuce, 8 targets, classical ANOVA	x̄ 4345.5625; s_T 556.2804, s_samp 518.16089, s_anal 148.18063; 51.58/44.76/3.66 %	same (s_samp 518.16087)	✅
UFS-A1R	same, robust ANOVA (RANOVA)	4408.3237; 565.40, 319.05, 167.94	robust mean by H15 on target means 4408.319; s not reproduced	❌ method unpublished (F63)

ID	Example	Published	Engine	Status
UFS-A2	lead in soil, 10 targets, classical and ln x	317.8; 197.55, 135.43, 17.99; U′ 85.23/11.32/85.98 %; s_G 0.66775, 0.4784, 0.0567, 0.4817; F_U 2.6032, 1.12, 2.6207; u′ 0.5111	same	✅
UFS-A3	dissolved iron, 6 wells, range method	errata: s_A 0.015, s_S 0.090, s_T 0.601 mg/L; U 1.8/10.5/70 %	same	✅ (F64)
UFS-A3A	same, classical ANOVA	U 1.6/9.6/70 %	same	✅
UFS-A4	vitamin A, 40 g test samples	SS 16595/14231; s_anal 28.805, s_samp 17.224; RSD_B 6.1 % (errata)	same	✅ (F65)
UFS-A4S	vitamin A, 4 g test samples	V_samp −2662.15 → 0; RSD_anal 36.68 %	same	✅
UFS-A4Q	duplicate control chart	CL 11, WL 27, AL 36 %; all D % below WL	10.88, 27.33, 35.56 %; 0 outside WL	✅ (F65)
UFS-A5	enzyme in feed, Gy	c 2160; C 540, 270; s_r 0.033, 0.13; s_t 0.143; U′ 26.8/28.6 %	2158.5; 539.6, 269.8; 0.0325, 0.1296; 0.1427; 26.7/28.5 %	✅ (F66)
UFS-A6	Cd and P in soil, modelling (GUM)	0.32 ± 0.06, 116 ± 26 mg/kg	0.319 ± 0.059, 115.8 ± 26.1	✅ (F67)
UFS-D2	unbalanced nested design	method of moments	core test: EMS coefficients and MS_anal	✅ (core)

Uncorrected bias (Magnusson & Ellison 2008) — core tests

ID	Claim	Published	Engine	Status
ME2008-T3	formulae of Table 3	RSSu, RSSU, SUMU, SUMU_MAX, Ue	same (SUMU signs per Phillips et al. 1997, F43)	✅
ME2008-Ue	Ue ≈ 1.65u_c + |Δ| for |Δ| > 0.8u_c	stated	within 3 % for |Δ|/u_c = 0.8 … 4	✅
ME2008-F1	coverage, u(y) = 1, u_Δ = 0.5	SUMU ≈ 95 % → 97.5 % at 3u_c; Ue ≈ 98 % → 95 %; RSSu, SUMU_MAX > 95 %	95.2/97.5 %; 98.1/95.0 %; > 95 %	✅
ME2008-F2	interval widths, u_Δ = 0.3	RSSU < Ue ≈ RSSu < SUMU_MAX for |b| ≤ u(y); RSSu largest at 2u(y)	same	✅

Measured MCM dispersion (σ over 20 seeds, M = 2 × 10⁶)

Case	y	u	shortest lo	shortest hi	shortest length	symmetric lo	symmetric hi
mass calibration [mg]	4.0e-5	3.7e-5	7.4e-4	7.4e-4	1.8e-4	1.0e-4	1.2e-4

Case	y	u	shortest lo	shortest hi	shortest length	symmetric lo	symmetric hi
loss $x_1 = 0$, r = 0 [10^{-6}]	0.04	0.04	6e-5	0.16	0.16	0.006	0.23
loss $x_1 = 0.01$, r = 0	0.10	0.07	4e-4	0.29	0.29	0.03	0.30
loss $x_1 = 0.05$, r = 0	0.48	0.20	3.2	3.3	1.0	0.78	0.85
loss $x_1 = 0$, r = 0.9	0.04	0.08	2e-5	0.20	0.20	0.003	0.30
loss $x_1 = 0.01$, r = 0.9	0.10	0.12	0.34	0.50	0.34	0.03	0.40
loss $x_1 = 0.05$, r = 0.9	0.48	0.23	5.0	4.9	1.0	0.77	0.97

Further examples (GUM-5, EMUE, IUPAC, consensus, qNMR and others)

ID	Example	Published	Engine	Status
EUR-CA-B1–B3	Eurachem/CITAC compliance guide 2021, Annex B Examples 1–3 and Table 1	B1 16.2–17.8 % Ni, non-compliant; B2 204.1 ng/g, compliant; B3 3.6 (lognormal) / 3.2 (normal) ng/g; Table 1 51/149, 61/164, 18/182, 44/227	16.164–17.836; 204.09; 3.557 / 3.151 (exact s_G 3.498); Table 1 with k = 1.6449 (227.6)	✅ except F86
EMUE-E3.4	EMUE E3.4, metals in soil (ISO 21748, ruggedness coefficients, zero-intercept calibration)	Tables E3.4.1–2; $u'(Pb)^2 = 0.002\,526$; (27.8 ± 2.8) mg/kg; calibration u_rel 0.019	same to rounding; RMS of $u(\hat{\delta})/x_{ref}$ 3.50 % (printed 3.3 %) → ± 2.9	✅ except F87
EMUE-E2.1	EMUE E2.1, total risks of a four-component medication	R*_tot 0.0029, 0.0117, 0.0002; R_tot 0.0018	0.002 88, 0.011 72, 0.000 21; 0.001 80 (program), 0.001 82 (numpy)	✅
EMUE-E2.2	EMUE E2.2, vacuum gauge: three scenarios for a reference correction; sum model with a common temperature error	Tables E2.2.2–E2.2.7 (30 p_c values); u(Δp) 1.14/0.91 Pa, p_c 0.67/0.75	all to 0.1 % (two rounding cases); 1.140/0.909, 0.669/0.745	✅ except F88
EMUE-E3.8	EMUE E3.8, NH_3 in N_2 by permeation	y = 70.8 µmol/mol, u = 0.599 µmol/mol; E_A 59.164 kJ/mol, k_0 43.069 µg/min, $\partial\dot{m}/\partial T$ $4.81 \cdot 10^{-7}$	70.796, 0.598; E_A 56.64 kJ/mol, k_0 $9.1 \cdot 10^9$, $\partial\dot{m}/\partial T$ 0.82 µg min^{-1} K^{-1}	✅ result; F89
EMUE-E3.3	EMUE E3.3, NO_2 analyser, dynamic dilution with correlated reference values	V_x (E3.3.17) diag 39.5/37.0/41.2, off 1.7/2.2/3.0; A 88.94, B 4.19, V_AB (E3.3.18); validation 982.0 ± 4.7 (4.5)	diag 1.34/2.53/4.48, off 1.76/2.32/3.19; V_AB exactly (program); A 89.21 (88.94 by full WTLS); 981.95 ± 4.52 (4.35)	✅ V_AB; F90
EMUE-E6.5	EMUE E6.5, ephedrine in urine (WADA top-down budget and decision limit)	u_c 3.8 %; DL 11 µg/mL; Table E6.5.2	u_c 3.89 %; DL 10.822 (rounded up to 11); means 11.83/11.67/11.47, z −0.93/−1.27/0.73 (σ = 0.5)	✅ except F91
IUPAC24-4A	IUPAC 2024 Example 4A, water hardness	u ≈ 3 mg/L	H 96.65, u 3.18	✅
IUPAC24-4B	IUPAC 2024 Example 4B, silicon isotopic abundances (3 outputs)	0.923 3873(36), 0.046 2300(22), 0.030 3827(30)	same	✅

ID	Example	Published	Engine	Status
IUPAC24-4H	IUPAC 2024 Example 4H, cadmium standard by MCM with beta purity and triangular volume	γ 1.0027, u 0.0009, U95 0.0017 mg/mL	GUM 1.002 70 (0.000 864); MCM 1.002 70 (0.000 864), (1.001 03, 1.004 37)	✅
IUPAC24-5C	IUPAC 2024 Example 5C, benzoic acid purity as beta(88 873.7, 7.11)	— (distribution given)	0.999 920 (0.000 030); 95 % (0.999 851, 0.999 968)	✅
IUPAC24-4I	IUPAC 2024 Example 4I, argentometric titration, conventional and K_sp models	10.000 ± 0.026 mmol/L; K_sp model differs by 0.24 %	10.000 ± 0.0255; 9.976 (mass balance)	✅ except F92
IUPAC24-3E-4M	IUPAC 2024 Examples 3E, 4M, 4N, molar mass of iridium: Q test, Bayesian random-effects model (MCMC in the guide)	p = 0.0082; M = 192.2171(4) g/mol, median τ 0.0005 g/mol	p = 0.0084; 192.217 119 (0.000 440), median τ 0.000 506 (exact grid)	✅ except F94
IUPAC24-4F	IUPAC 2024 Example 4F, CCQM-K145 Ni in bovine liver, maximum likelihood with dark uncertainty	$\hat{\mu}$ 2.043(15), $\hat{\tau}$ 0.044(14) mg/kg	2.042 94 (0.015 44), 0.043 54 (0.014 24)	✅
KOEPKE17-G	Koepke et al. 2017 Tables 1–2, Newtonian constant of gravitation (DL, bootstrap, Linear Pool, ML)	DL 6.673 80, 0.000 28, (6.673 20, 6.674 40); LP 6.673 67, 0.001 24, (6.671 18, 6.675 79); ML 6.673 78, 0.000 30	DL 6.673 790, 0.000 279, (6.673 23, 6.674 35); LP 6.673 667, 0.001 243, (6.671 17, 6.675 79); ML 6.673 778, 0.000 300	✅ except F93
KOEPKE17-BAYES	Koepke et al. 2017 Tables 2 and 4, hierarchical Bayesian rows (NICOB §A.3, σ_j = u_j)	G 6.673 77, 0.000 33, (6.673 10, 6.674 42); ^{137}Cs 29.94 (0.19) a; ^{90}Sr 28.71 (0.16) a	6.673 773, 0.000 329, (6.673 105, 6.674 409); 29.944 (0.185) a; 28.708 (0.160) a	✅ except F95
PM22-HUBBLE	Possolo & Meija 2022, Hubble–Lemaître constant, Bayesian consensus with Laplace effects (Stan)	71.69, 0.66, median τ 2.03 (km/s)/Mpc	71.6915, 0.6603, 2.0336 (grid)	✅
PM22-AG	Possolo & Meija 2022, plasma anion gap	25, u 1.85 mmol/L	25, 1.851	✅
PM22-TANK	Possolo & Meija 2022, storage tank V = πR²H	7204 ± 54 m³ (Gauss and MCM)	7204.3, 54.0 (both)	✅
PM22-WB	Possolo & Meija 2022, Wheatstone bridge	957 Ω, u ≈ 7 Ω; E 52 %, F 48 %, G 0.6 %, H 0.004 %	957.2, 6.98; same shares	✅
PM22-PB	Possolo & Meija 2022, atomic weight of lead in HIPB-1 (4 amount fractions and A_r)	u(x) 6, 37, 19, 45 × 10^{-6}; r −0.02, +0.13, +0.18, −0.17, −0.90, −0.58; A_r 207.1791, U95 0.0002	same u(x); r −0.011, 0.124, 0.177, −0.179, −0.895, −0.576; 207.17915, U95 0.000 17	✅ except F96
BIPM-ISRD-qNMR	BIPM ISRD-01, -02, -06, -07, -08: nine ^{1}H qNMR purity budgets	w_A and u_c of each table (e.g. ISRD-01 T4 0.9973, 7.30 × 10^{-4})	same w_A to rounding; u_c within 1 % except ISRD-01 T6 (4.10 vs 3.96 × 10^{-4})	✅ except F97
PB18-CR	Possolo & Bodnar 2018 §4.4, chromium in soil (data of ISO Guide 35:2017 C.1, now ISO 33405:2024), ANOVA and hierarchical Bayes	REML 121.62 (0.95), τ 3.93, σ 2.87 mg/kg; MCMC 121.62 (1) mg/kg	121.624; s_bb 3.930, s_r 2.874; Bayes grid 121.624 (0.9986)	✅
MERKATAS19-G	Merkatas et al. 2019 Table 4, G from 16 results: DL (Knapp–Hartung), ML, NICOB Bayes, Linear Pool	DL 6.673 99 (0.000 25), τ 0.000 56; ML 6.673 90 (0.000 25); BG 6.673 89 (0.000 27); LOP 6.673 77 (0.001 17)	6.673 993 (0.000 250), 0.000 556; 6.673 900 (0.000 252); 6.673 896 (0.000 271); 6.673 773 (0.001 168)	✅

ID	Example	Published	Engine	Status
UME-PROC-19	TÜBİTAK UME draft procedure, Examples 19.1 (DMM 10 V) and 19.2 (qNMR purity, illustrative)	19.1 u_c 35.1 µV, ν_eff ≈ 4000, U 70.2 µV; 19.2 995.0 mg/g, u_c 1.567, ν_eff 121, U 3.13 mg/g	35.09, 3965, 70.19; 995.00, 1.5669, 121.5, 3.134	✅
TUR-GB	TUR guard bands (Dobbert 2.5, RSS 1.5, RP-10 2.0) with a 95 % in-tolerance normal process	published formulas and JCGM 106 9.5 integrals in SciPy (D110; Suncal 1.7.1 agrees): GBF 0.914 186, 0.745 356, 0.75; PFA 0.6883 %, 0.5679 %, 0.3213 %; PFR 5.216 %, 17.70 %, 14.00 %	same to 2×10^{-6}	✅
CG18-H3.4	EURAMET cg-18 H3.4/A and /B, 30 t weighbridge in use (substitution loads, drift, ΔT = 40 K)	α^2 62.133 / 25.333 kg²; β^2 1.276×10^{-6} / 1.960×10^{-7}; U = 16 kg + 1.79×10^{-3} R / 10.067 kg + 6.113×10^{-4} R; R_min 2169 / 1123 kg	62.133 / 25.333; 1.276×10^{-6} / 1.967×10^{-7}; 15.76 + 1.794×10^{-3} R / 10.066 + 6.128×10^{-4} R; 2169 / 1124	✅ except F98
KOEPKE17-CS	Koepke et al. 2017 Tables 3–4, half-life of ^{137}Cs (19 results)	DL 30.04 (0.04) a; LP 29.94 (0.93) a	30.037 (0.045) a; 29.94 (0.93) a	✅
KOEPKE17-SR	Koepke et al. 2017 Tables 3–4, half-life of ^{90}Sr (11 results)	DL 28.73 (0.11) a; LP 28.68 (0.77) a	28.730 (0.110) a; 28.69 (0.77) a	✅
NICOB-CO60	NIST Consensus Builder manual §4.4, ^{60}Co, 19 laboratories: DL bootstrap, Linear Pool, Bayes; leave-one-out and bilateral DoE of DL, LP and Bayes (numpy port of the NICOB source, 10^6 draws)	DL 7062, u 4, (7053, 7071) kBq; LP 7064, 29, (7012, 7127); Bayes 7062, 5, (7053, 7072); DoE: D exact (LP: within the MC error of the pool mean), U95 from the port	7062.06, 4.33, (7053.4, 7070.7); 7063.8, 29.2, (7012.6, 7127.2); 7062.03, 4.59, (7053.1, 7071.4); U95 within 1.3 %; significant pairs as the port	✅
NICOB-PCB28	NIST Consensus Builder manual §4.2, PCB 28, 6 laboratories with ν (one ν = 2)	DL 33.6, u 0.77, (32.0, 35.2) ng/g; LP 33.6, 1.55, (31.4, 36.3)	33.600, 0.751, (32.06, 35.15); 33.64, 1.51, (31.44, 36.28); DoE U95 within 1.7 % of the port	✅ except F99

Example library (project files)

All cases of this matrix are stored as self-checking project files in `library/` (286 projects; supplementary material S1), run by `project/tests/library.rs` and `kapsam check library` through the complete path a user takes (JSON → evaluation → comparison → TR/EN reports). Reference-value policy: D15 and D110 in the research log. EA-4/02 S13 in the library uses the raw observations (estimate 0.2323 µm, F21), whereas the core test reproduces the printed budget. The 39 top-down projects start from the raw data wherever the documents give them (duplicate pairs, control-chart results, replicate groups, PT results); the two projects from Bettencourt da Silva (2026) start from the 20 mean recoveries of the spreadsheet (`"type": "given"`). Tolerances reflect the documents' printed rounding; where a guide computes from rounded intermediates the tolerance is widened and the reason stated in the file.

Not yet covered

TN 1900 observation-equation examples (regression, random effects); TN 1900 E5 (truncated Gaussian inputs); the data sets of ISO/TS 28037 (the generalised least-squares engine of ISO 6143 covers the method); the allowance of MUPR Appendix C as a project file.

Values close to their tolerance

Generated by `tools/research/article_stats.py`. Every compared value with |Δ| > 0.9 × tolerance. Where the tolerance is half a unit in the last printed digit, the exact value lies near the rounding limit of the printed one (e.g. 93.155 printed as 93.16); the other tolerances are wider for the reason stated in the project file.

74 of 1865 values; 51 of them with a tolerance of half a unit in the last digit.

Project	Value	Reference	Computed	Tolerance	|Δ|/tol	Basis of the tolerance
m3003-k2-power-sensor-18ghz	gum.estimate	93.16	93.155	0.005	1.000	half a unit in the last digit
iso5725-2-c2-softening-point-pitch	precision_experiment.levels[3].cochran	0.434	0.4335022166	0.0005	0.996	half a unit in the last digit
iso5725-2-c3-creosote-oil	precision_experiment.levels[2].grubbs_double_high	0.395	0.3945023545	0.0005	0.995	half a unit in the last digit
iso5725-2-c2-softening-point-pitch	precision_experiment.levels[4].grubbs_high	1.74	1.735027198	0.005	0.995	half a unit in the last digit
tr569-table1-zinc	control_chart.lower_action	52.48	52.48496751	0.005	0.994	half a unit in the last digit
iso5725-2-c1-sulfur-in-coal	precision_experiment.levels[2].grubbs_high	2.08	2.084960742	0.005	0.992	half a unit in the last digit
num-3-ratio-abc	mcm.standard_uncertainty	13	12.50427982	0.5	0.991	Monte Carlo dispersion
m3003-k3-attenuator-30db	gum.expanded_uncertainty	0.0491	0.04914943201	5e-05	0.989	half a unit in the last digit
jcgm101-9.4-comparison-loss-x050	gum.standard_uncertainty	502	502.4937811	0.5	0.988	half a unit in the last digit
iso21748-c4-crude-fibre	topdown.level[2.3].standard_uncertainty	0.31	0.3149322679	0.005	0.986	half a unit in the last digit
nist-tn1900-e2-surface-temperature	gum.standard_uncertainty	0.872	0.872491062	0.0005	0.982	half a unit in the last digit
tr569-ex8-copper-x-review	control_chart.upper_action	1.26	1.2551	0.005	0.980	half a unit in the last digit
tr569-ex8-copper-x-review	control_chart.lower_action	0.85	0.8549	0.005	0.980	half a unit in the last digit
iso13528-e4-mercury-feed	proficiency.scores[L09].z_prime	-3.47	-3.474986936	0.0051	0.978	up to one unit in the last digit
jcgm102-9.4-resistance-reactance-alternative	multivariate.gum.u[Z]	0.431	0.4314887653	0.0005	0.978	half a unit in the last digit
mupr-b4-arsenic-one-crm	topdown.recovery.statistic	2.24	2.235115946	0.005	0.977	half a unit in the last digit
gum-h2-reactance	gum.estimate	219.847	219.8465119	0.0005	0.976	half a unit in the last digit
jcgm102-9.4-resistance-reactance-alternative	multivariate.gum.estimate[X]	219.847	219.8465119	0.0005	0.976	half a unit in the last digit
jcgm102-9.4-resistance-reactance	multivariate.gum.estimate[X]	219.847	219.8465119	0.0005	0.976	half a unit in the last digit
koepke2017-half-life-cs137	consensus.u	14.61	16.3951656	1.83	0.976	wider (justified in the project's reference or notes)
cg18-h3-30t-10t-weights	gum.expanded_uncertainty	19	18.51300187	0.5	0.974	half a unit in the last digit

Project	Value	Reference	Computed	Tolerance	|Δ|/tol	Basis of the tolerance
cg18-h1-100g-option2	gum.expanded_uncertainty	0.00033	0.000334855096	5e-06	0.971	half a unit in the last digit
iso13528-e4-mercury-feed	proficiency.scores[L10].pa	5.1	5.050505051	0.051	0.970	up to one unit in the last digit
iso13528-e4-mercury-feed	proficiency.scores[L24].pa	5.1	5.050505051	0.051	0.970	up to one unit in the last digit
cg18-h4-400g-error-function-sm-0.05	error_function.u_approximated[4]	0.047	0.04748524296	0.0005	0.970	half a unit in the last digit
gum-h2-reactance	gum.standard_uncertainty	0.295	0.2955816552	0.0006	0.969	up to one unit in the last digit
jcgm101-9.3-mass-calibration	gum.standard_uncertainty	0.0539	0.05385164807	5e-05	0.967	half a unit in the last digit
nordtest-8-bod-pt	topdown.component[u(bias)]	4.12	4.134490778	0.015	0.966	wider (justified in the project's reference or notes)
possolo-meija-2022-atomic-weight-lead-hipb1	multivariate.gum.estimate[ArPb]	207.1791	207.1791483	5e-05	0.966	half a unit in the last digit
cg18-h2-60kg-10kg-option1	gum.standard_uncertainty	1.552	1.552481884	0.0005	0.964	half a unit in the last digit
mupr-b1-nitrate	topdown.recovery.standard_uncertainty_percent	2.85	2.854801397	0.005	0.960	half a unit in the last digit
cg18-h2-60kg-max-adjusted	gum.standard_uncertainty	5.739	5.724599695	0.015	0.960	wider (justified in the project's reference or notes)
jcgm102-9.5.3-ten-temperatures	multivariate.gum.u[theta2]	0.0027	0.002747846836	5e-05	0.957	half a unit in the last digit
ufs-a5-enzyme-feed-gy	sampling.gy.rsd[0]	0.033	0.03252183882	0.0005	0.956	half a unit in the last digit
cg18-h3-30t-10t-weights	gum.effective_dof	9	9.810291376	0.85	0.953	up to one unit in the last digit
gum5-e1-ph-temperature-interpolation-correlated	gum.estimate	7.0043	7.004347662	5e-05	0.953	half a unit in the last digit
gum5-e1-ph-temperature-interpolation	gum.estimate	7.0043	7.004347662	5e-05	0.953	half a unit in the last digit
iso5725-2-c3-creosote-oil	precision_experiment.levels[5].s_r	0.393	0.3934735424	0.0005	0.947	half a unit in the last digit
iso5725-2-c3-creosote-oil-reml	precision_experiment.levels[5].s_r	0.393	0.3934735399	0.0005	0.947	half a unit in the last digit
gum5-e5-ghg-inventory-agriculture-shared-activity	gum.estimate	4282.04	4282.144091	0.11	0.946	wider (justified in the project's reference or notes)
gum5-e5-ghg-inventory-agriculture	gum.estimate	4282.04	4282.144091	0.11	0.946	wider (justified in the project's reference or notes)
koepke2017-newtonian-constant-g	consensus.mean	6.6738	6.673789609	1.1e-05	0.945	half a unit in the last digit
iso13528-e4-mercury-feed	proficiency.scores[L07].z_prime	-0.51	-0.5148128794	0.0051	0.944	up to one unit in the last digit
iso13528-e4-mercury-feed	proficiency.scores[L21].z_prime	-0.51	-0.5148128794	0.0051	0.944	up to one unit in the last digit
iso13528-e4-mercury-feed	proficiency.scores[L25].z_prime	-0.51	-0.5148128794	0.0051	0.944	up to one unit in the last digit
tr569-ex12-pooling-sr-srw	control_chart.precision.s_rw_simplified	0.27	0.2704713698	0.0005	0.943	half a unit in the last digit

Project	Value	Reference	Computed	Tolerance	|Δ|/tol	Basis of the tolerance
nordtest-3-ammonium-pt	topdown.component[u(bias)]	2.73	2.725288909	0.005	0.942	half a unit in the last digit
ea402-s7-step-attenuator-30db	gum.expanded_uncertainty	0.045	0.04443891386	0.0006	0.935	up to one unit in the last digit
iso13528-e4-mercury-feed	proficiency.scores[L01].en	0.83	0.8347659171	0.0051	0.934	up to one unit in the last digit
cg18-h2-60kg-max-option1	gum.expanded_uncertainty	12.254	12.11409702	0.15	0.933	wider (justified in the project's reference or notes)
nordtest-5.3-oxygen-duplicates	topdown.component[s_r, duplicates]	0.34	0.3353377891	0.005	0.932	half a unit in the last digit
iso13528-e4-mercury-feed	proficiency.robust_sd	0.0164	0.01644655431	5e-05	0.931	half a unit in the last digit
iso13528-e4-mercury-feed	proficiency.scores[L29].pa	-25.3	-25.25252525	0.051	0.931	up to one unit in the last digit
iso13528-e4-mercury-feed	proficiency.scores[L28].pa	25.3	25.25252525	0.051	0.931	up to one unit in the last digit
mupr-b6-sediment-22-pt	topdown.recovery.mean_percent	91.735	91.7345348	0.0005	0.930	half a unit in the last digit
jcgm102-9.5.3-ten-temperatures	multivariate.gum.r(theta8, theta10)	0.549	0.5494635718	0.0005	0.927	half a unit in the last digit
m3003-k10-sheet-thickness	gum.effective_dof	50	50.92335562	1	0.923	half a unit in the last digit
emue-e2.2-vacuum-gauge-correction-ignored	gum.estimate	1.002	1.001953888	5e-05	0.922	half a unit in the last digit
emue-e2.2-vacuum-gauge-correction-in-uncertainty	gum.estimate	1.002	1.001953888	5e-05	0.922	half a unit in the last digit
iso13528-e4-mercury-feed	proficiency.scores[L05].en	-2.88	-2.875304825	0.0051	0.921	up to one unit in the last digit
tr569-ex8-copper-x-new-limits	control_chart.upper_action	1.29	1.2946	0.005	0.920	half a unit in the last digit
iupac2024-4f-ccqm-k145-nickel-bovine-liver	consensus.ml_tau	0.044	0.04354104957	0.0005	0.918	half a unit in the last digit
mupr-b6-sediment-22-pt	topdown.recovery.chi_squared	8.7	8.70458337	0.005	0.917	half a unit in the last digit
ea402-s4-gauge-block-50mm	gum.expanded_uncertainty	69	68.54214438	0.5	0.916	half a unit in the last digit
tr569-ex6-phosphorus-relative-range	control_chart.upper_action	6.16	6.155428229	0.005	0.914	half a unit in the last digit
iso13528-e4-mercury-feed	proficiency.scores[L03].pa	-35.4	-35.35353535	0.051	0.911	up to one unit in the last digit
gum5-e9-torque-slope-ols	gum.estimate	1.0107	1.01065447	5e-05	0.911	half a unit in the last digit
quam-a4-pesticides-topdown	topdown.recovery.statistic	2.31	2.314550249	0.005	0.910	half a unit in the last digit
iso5725-2-c2-softening-point-pitch-reml	precision_experiment.levels[4].s_R	1.918	1.917545115	0.0005	0.910	half a unit in the last digit
iso5725-2-c1-sulfur-in-coal-reml	precision_experiment.levels[2].s_R	0.062	0.06154627493	0.0005	0.907	half a unit in the last digit
nordtest-8-bod-pt	topdown.level[relative].relative_standard_percent	4.87	4.883589591	0.015	0.906	wider (justified in the project's reference or notes)

Project	Value	Reference	Computed	Tolerance	|Δ|/tol	Basis of the tolerance
nordtest-8-bod-pt	topdown.level[relative].relative_expanded_percent	9.74	9.767179183	0.03	0.906	wider (justified in the project's reference or notes)
emue-e2.2-vacuum-gauge-correction-applied	gum.estimate	1.0025	1.002454865	5e-05	0.903	half a unit in the last digit
cg18-h4-400g-error-function-sm-0.25	error_function.u_approximated[4]	0.084	0.08354975691	0.0005	0.900	half a unit in the last digit

Monte Carlo results under other seeds

Written by `tools/research/mc_seeds.py`. Each of the 61 library projects with a Monte Carlo check was run with the ten seeds 101–110 instead of its own, and every Monte Carlo result was compared with its reference value and tolerance.

- comparisons: 2810 (281 Monte Carlo values × 10 seeds); runs that failed: 0
- |Δ|/tolerance: median 0.178, 90 % 0.608, 99 % 0.896, maximum 0.981
- outside the tolerance: 0

History (D110): the first run of this test found 20 comparisons from 7 values outside their tolerance (at most 1.26 times it): GUM-5 E14 and E15 (estimate), NUM §3 (lower interval end) and §8 (estimate), TN 1900 E8 (upper interval end) and the correlations r(R, X) and r(R, Z) of JCGM 102 9.4 (two projects). Their tolerances had been set from the program's own seed; they were reset to half a unit in the last printed digit plus five times the standard deviation over eleven seeds, and the run above was repeated.